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AI Architecture & Digital Twin Engineering Portfolio

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National Higher School of Computer Science (ESI-SBA)

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Agenda

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- 4 Medical Digital Twin (Master's Thesis)
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- 6 Study Program – IASD
- 7 Emerging AI Technologies
- 8 Full-Stack Engineering

Who Am I?

- **Profile:** AI & Data Science Engineer graduating in 2026.
- **Education:** State Engineering Degree & Master's Degree in Computer Science from ESI-SBA.
- **Engineering Philosophy:** Moving beyond "Black-Box" AI. My work focuses on architecting systems where neural networks are constrained by real-world physical laws and logic.
- **Core Domains:**
 - Physics-Informed Neural Networks (PINNs)
 - Spatio-Temporal Sequence Modeling (Mamba, Transformers)
 - Graph Neural Networks & Reinforcement Learning
 - Full-Stack System Architecture (FastAPI, Next.js, PostgreSQL)

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2021 – 2026

- **Machine Learning:** Regression, SVMs, Decision Trees, Ensembles (Random Forest, XGBoost), Dimensionality Reduction (PCA, t-SNE).
- **Deep Learning:** CNNs (ResNet, EfficientNet), RNNs/LSTMs, Autoencoders, GANs, Physics-Informed Neural Networks (PINNs).
- **Computer Vision:** Object Detection (YOLO), Semantic Segmentation (U-Net), Vision Transformers (ViT).
- **Natural Language Processing (NLP):** Word Embeddings (Word2Vec), Attention Mechanisms, Transformers (BERT, GPT), RAG architectures.
- **Advanced AI & Big Data:** Graph Neural Networks (GNNs), Deep Reinforcement Learning (DQN, PPO), Big Data ecosystems (Hadoop, Spark), Semantic Web (Knowledge Graphs).

- **DeepLearning.AI (Coursera)**

- Machine Learning Specialization
- Deep Learning Specialization
- Natural Language Processing (NLP) Specialization

- **Zero To Mastery (ZTM) Academy**

- PyTorch for Deep Learning: Zero to Mastery
- TensorFlow Developer Certificate Bootcamp

Engineering Thesis: Gas Turbine Digital Twin

Context: Real-time monitoring of GE LM2500 gas turbines for Sonatrach.

Challenge: Hardware Latency Paradox (Cloud AI is too slow for emergency shutdowns, edge hardware cannot process massive models).

Solution: A "Dual-Track" 3-Tier Pipeline decoupling the logic.

- **TCNEdge (Temporal Convolutional Network):** 1D CNNs optimized for the Edge. Acts as an immediate safety trigger, bypassing complex attention mechanisms for pure speed (0.049 ms inference).
- **MambaTS (Selective State Space Model):** A near-edge sequence model providing linear $O(N)$ complexity for infinite-horizon memory, enhanced with a "missingness-aware gate" for sensor faults.
- **ST-KDFormer & KAN-PCNN (Cloud):** Spatio-Temporal Knowledge-Distilled Transformers map complex interactions, while Kolmogorov-Arnold Networks act as Physics-Constrained NNs to obey thermodynamics.

Context: Predicting sudden drops in Mean Blood Pressure (MBP) in Intensive Care.

Challenge: Models overfit to time-series noise and ignore fundamental fluid dynamics.

Data: 100Hz VitalDB dataset with strict 70/15/15 patient-level split.

- **PI_DeepONet (Physics-Informed Deep Operator Network):** A dual-network architecture (Branch Net for patient anatomy, Trunk Net for time). It learns mathematical operators (PDEs) rather than just functions.
- **0D Windkessel Model:** Acts as the physics engine. PyTorch Autograd calculates derivatives to penalize predictions violating physical vascular resistance or arterial compliance.
- **Medical_MambaTS:** Deployed at the bedside for continuous state-space monitoring without memory bottlenecks.

Architecture: Bridging Sequence Learning and Reinforcement Learning

- **LSTM (Long Short-Term Memory):** A specialized Recurrent Neural Network used in the Prediction Layer. It processes historical market data to capture long-term temporal dependencies and predict short-term price movements.
- **DQN (Deep Q-Network):** Used in the Action Layer. A Reinforcement Learning algorithm that combines Q-Learning with deep neural networks. It takes LSTM outputs as states to optimize long-term trading reward policies.
- **Infrastructure:** Highly scalable **FastAPI** backend and a reactive **Next.js** dashboard receiving real-time agent telemetry via WebSockets.

Architecture: Graph-Based Recommendation System

- **SBERT (Sentence-BERT):** A modification of the pretrained BERT network used to derive semantically meaningful sentence embeddings. It reads resumes and job descriptions to generate deep semantic vectors.
- **GNN (Graph Neural Network):** A class of neural networks designed to process data represented as graphs (using PyTorch Geometric). It processes the SBERT embeddings as nodes, learning the hidden, non-linear relational patterns between a candidate's skills and a job's true requirements.

Elydent & Tajer (Professional Desktop POS Systems)

- **The Philosophy:** AI models are useless if they cannot be delivered in robust, usable software. I build production-ready platforms.
- **How they are Built:**
 - **Architecture:** Packaged as cross-platform desktop applications using **Electron**.
 - **Frontend:** Built with **Next.js 14** and **Tailwind CSS** for highly responsive, modern UI/UX.
 - **Database:** Implemented robust, offline-first **SQLite** databases with complex relational structures for inventory, patient tracking, and accounting.

Thank You!

Questions?